Should a move's verdict follow the mover's rating?

A rating-aware win-probability curve, put to a blind judge

Chess Coach research lab

Abstract

The coaching app labels each move (good, ok, inaccuracy, mistake, blunder) from how much of the mover's winning chances it gives away, read on the win-probability curve Lichess uses. A natural objection is that the curve is calibrated on stronger players: near a level position, a pawn decides fewer games between lower-rated players. We tested a proposal that flattens the curve for lower-rated movers, with a slope of 0.00245 per centipawn at or below 1000 rising linearly to the shipped 0.00368208 at 2300. Over 60 of the player's games, the proposal changes 492 of the 3578 graded verdicts (13.8%). A judge model, shown 25 of the player's own flipped moves with the two labels in random order and unnamed, preferred the shipped label 13 and 15 times in two runs, and the proposed one 12 and 10 times. The proposal moves one verdict in seven without making verdicts fairer on balance, and was not adopted. The judge's reasons point elsewhere: to what a move gives away (material, a mate) rather than to the size of the swing.

1 The question

A game review is only useful if its verdicts match what the player can learn from. The app's judge reads the mover's win probability before and after the move and labels the move by the drop. The curve that turns an engine evaluation into a win probability is the one Lichess fits on its own games, and it does not depend on who is playing. For a player at this level, a half-pawn concession in a level position rarely decides the game, so a curve for that level might be flatter near zero, and verdicts there milder.

The question here is whether such a curve gives fairer verdicts *to this player*: an adult club player, rated 948–1167 on chess.com across the 60 games studied here, 59 of them played at 10 minutes a side.

2 The judge and the proposal

The win probability of the mover, for an evaluation of x centipawns from the mover's side, is

$$W_k(x) \;=\; 50 + 50\left(\frac{2}{1+e^{-kx}} - 1\right),$$

with $k = 0.00368208$ in the app. The judge takes the drop $\Delta = W_k(x_{\text{before}}) - W_k(x_{\text{after}})$ in percentage points and labels the move a blunder at $\Delta \geq 20$, a mistake at ≥ 10, an inaccuracy at ≥ 5, good at ≤ 1, and ok otherwise. Two clamps protect a decided game: if the mover still has at least 88% after the move, any label but good becomes ok, and at 75% or more, a blunder or mistake becomes an inaccuracy. The judge reads ok and good alike as a good move, so a flip between them does not change the label the player sees.

The proposal changes only the slope, by the mover's rating r:

$$k(r) \;=\; 0.00245 + (0.00368208 - 0.00245) \cdot \mathrm{clamp}\left(\frac{r - 1000}{1300},\, 0,\, 1\right).$$

Thresholds and clamps stay as shipped. Figure 1 shows the two curves for a mover at or below 1000.

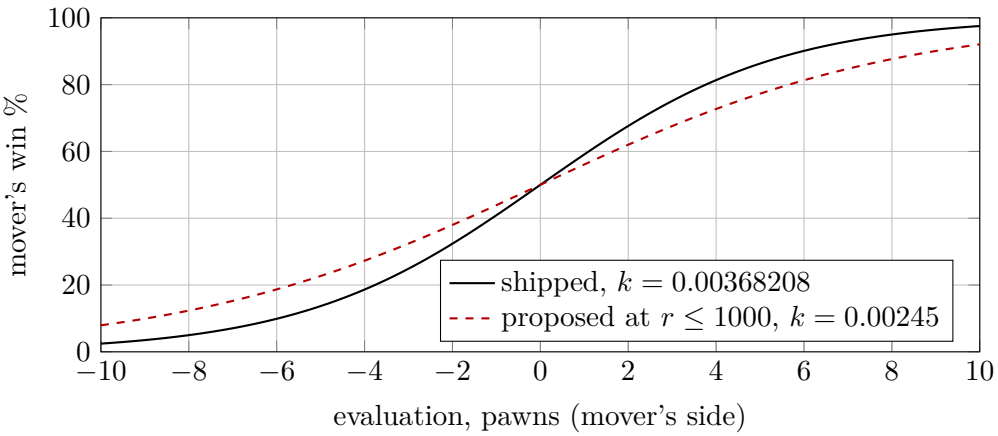


Figure 1: The two win-probability curves. Near level the proposed curve counts each pawn as fewer win points, so a drop there reads milder; further out it counts each pawn as more.

Where the curves part. The two curves change win probability at the same rate per pawn at about 5.1 pawns from level on either side. Nearer level, the proposed curve moves less per pawn and its verdicts are milder; further out it moves more and they are harsher. The clamps move too: the 88% floor needs about +5.4 pawns on the shipped curve and +8.1 on the proposed one, and the 75% cap needs +3.0 and +4.5. A move that leaves the mover between those levels is protected on the shipped curve and not on the proposed one.

3 How many verdicts change

The data are the phone-profile reviews of 60 of the player's games (3917 moves, 3880 with an exact engine score before and after; the other 37 end the game with mate, which the outcome grades). Book and forced moves (254 and 48) are not graded, which leaves 3578 graded moves, 1783 by the player and 1795 by the opponents. Player ratings in the set run from 948 to 1167 and opponent ratings from 513 to 1178, so nearly every mover sits at or near the proposal's floor.

Every graded move is judged twice by a copy of the app's judge that takes the slope as a parameter. On the shipped slope the copy reproduces the app's own judge on all 3880 exact-score moves, with the largest win-probability difference 1.4×10^{-14}, so the comparison isolates the curve.

The proposal changes 492 verdicts (13.8% of graded moves): 250 of the player's (14.0%) and 242 of the opponents' (13.5%). Table 1 shows where they go.

Most flips are milder near level (an inaccuracy that becomes ok, a mistake that becomes an inaccuracy, a blunder that becomes a mistake). The harsher ones come from positions already far from level, where the proposed curve moves more per pawn and its clamps protect less.

shipped ↓ proposed →	blunder	mistake	inaccuracy	ok	good	total
blunder	(37)	**29**	0	0	0	66
mistake	**4**	(50)	**48**	0	0	102
inaccuracy	**1**	**9**	(76)	**54**	0	140
ok	0	0	**16**	(439)	**78**	533
good	0	0	0	**11**	(931)	942
total	42	88	140	504	1009	1783

(a) The player's moves.

shipped ↓ proposed →	blunder	mistake	inaccuracy	ok	good	total
blunder	(52)	**28**	0	0	0	80
mistake	**2**	(45)	**50**	0	0	97
inaccuracy	0	**10**	(75)	**47**	0	132
ok	0	**2**	**12**	(500)	**76**	590
good	0	0	0	**15**	(881)	896
total	54	85	137	562	957	1795

(b) The opponents' moves.

Table 1: Verdicts on the shipped curve (rows) and on the proposed one (columns). Unchanged verdicts are in parentheses; flips are in bold.

4 What a judge preferred

Protocol. The 25 moves are the player's own flipped moves whose shown label changes (the moves on the app's Coach Calibration page). For each, a judge model was given the position before the move, the move, the engine's evaluation with the best move and after the move played, the engine's best move and line, both players' ratings, and the two labels as X and Y in random order, unnamed. It was asked, as an experienced coach whose student is "an adult club player rated about 950–1150 on chess.com who plays 10-minute games", which label matches how bad the move really was at this level, or to say it could not tell, with one sentence of reason. It answered twice, independently, with the order of the labels reversed.

Result. Table 2 shows the preferences. The two runs gave the same answer on 23 of 25 moves. Neither run prefers the proposed curve.

By kind, the judge took the proposed side where a small concession near level becomes a good move, and kept the shipped side where the move simply hangs material, where a piece is trapped, and where the game was already decided and the proposal would raise the verdict. Figures 2–5 show one move of each of the four commonest kinds, with the move played (red arrow) and the engine's best move (green).

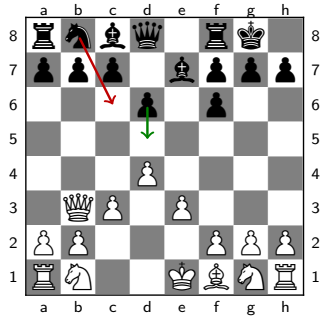
	run 1	run 2
shipped verdict preferred	13	15
proposed verdict preferred	12	10
unsure	0	0

(a) Each run.

shown label: shipped → proposed	moves	both runs: shipped	both runs: proposed	split
Inaccuracy → Good	7	2	5	0
Mistake → Inaccuracy	7	4	1	2
Blunder → Mistake	4	2	2	0
Good → Inaccuracy	2	2	0	0
Inaccuracy → Mistake	2	1	1	0
Mistake → Blunder	2	1	1	0
Inaccuracy → Blunder	1	1	0	0

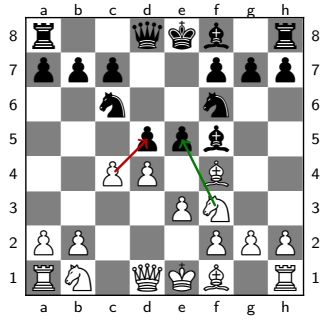
(b) By the kind of change, as the player sees the labels.

Table 2: Which verdict the judge preferred.



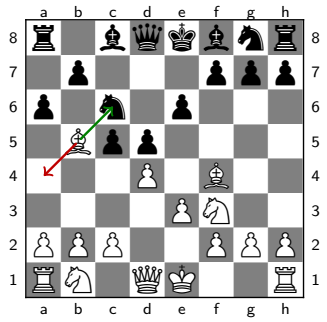
Played: 6. . . ♘c6, Black to move; ratings 1076 (the player) and 1074.
Engine: +0.29 with the best move, −0.33 after yours.
Best line: 6. . . d5 7. ♘d2 a5 8. h4 a4 9. ♕c2
UCI: `b8c6`; best line `d6d5 b1d2 a7a5 h2h4 a5a4 b3c2`
Shipped: Inaccuracy. **Proposed:** Good. **Judge:** the proposed verdict, in both runs.
Reason (run 1): "Nc6 is a normal developing move and the position stays balanced; the engine's small preference for ...d5 does not matter at this level."
FEN: `rnbq1rk1/ppp1bppp/3p1p2/8/3P4/1QP1P3/PP3PPP/RN2KBNR b KQ - 0 6`

Figure 2: Inaccuracy on the shipped curve, Good on the proposed one: the judge preferred the proposed label in both runs.



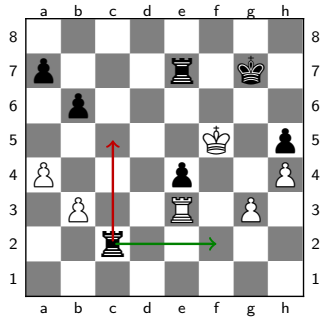
Played: 6. cxd5, White to move; ratings 1040 (the player) and 1052.
Engine: +1.05 with the best move, −0.35 after yours.
Best line: 6. ♘xe5 ♘b4 7. ♘a3 c6 8. ♗e2 ♗d6
UCI: `c4d5`; best line `f3e5 c6b4 b1a3 c7c6 f1e2 f8d6`
Shipped: Mistake. **Proposed:** Inaccuracy. **Judge:** the shipped verdict, in both runs.
Reason (run 1): "Black's e5 pawn was simply hanging to Nxe5, and cxd5 instead allows ...exf4/...Bb4+ play where White is no longer better, so a full pawn's worth is thrown away."
FEN: `r2qkb1r/ppp2ppp/2n2n2/3ppb2/2PP1B2/4PN2/PP3PPP/RN1QKB1R w KQkq - 0 6`

Figure 3: Mistake on the shipped curve, Inaccuracy on the proposed one: the judge preferred the shipped label in both runs.



Played: 6. ♗a4, White to move; ratings 1102 (the player) and 1105.
Engine: +0.31 with the best move, −2.51 after yours.
Best line: 6. ♗xc6+ bxc6 7. ♘c3 cxd4 8. exd4 f6
UCI: `b5a4`; best line `b5c6 b7c6 b1c3 c5d4 e3d4 f7f6`
Shipped: Blunder. **Proposed:** Mistake. **Judge:** the shipped verdict, in both runs.
Reason (run 1): "Ba4 walks into ...b5 and ...c4, which traps the bishop and costs a piece for a pawn: decisive material in the opening."
FEN: `r1bqkbnr/1p3ppp/p1n1p3/1Bpp4/3P1B2/4PN2/PPP2PPP/RN1QK2R w KQkq - 0 6`

Figure 4: Blunder on the shipped curve, Mistake on the proposed one: the judge preferred the shipped label in both runs.



Played: 45. . . ♖c5+, Black to move; ratings 1105 (the player) and 1099.
Engine: you mate in 3 with the best move, +5.91 after yours.
Best line: 45. . . ♖f2+ 46. ♖f3 ♖xf3+ 47. ♔g5 ♖e5#
UCI: `c2c5`; best line `c2f2 e3f3 f2f3 f5g5 e7e5`
Shipped: Good. **Proposed:** Inaccuracy. **Judge:** the shipped verdict, in both runs.
Reason (run 1): "Rf2+ mated in three, but after Rc5+ Black is still a full rook up and easily winning, so nothing important was given away."
FEN: `8/p3r1k1/1p6/5K1p/P3p2P/1P2R1P1/2r5/8 b - - 1 45`

Figure 5: Good on the shipped curve, Inaccuracy on the proposed one: the judge preferred the shipped label in both runs.

5 Decision, and what the reasons suggest

The proposal would move one verdict in seven, and in neither run did the judge prefer the proposed verdict more often than the shipped one, so it was not adopted. The reasons the judge gives are about consequences, not the size of the swing: a move that hangs material near level is a mistake whatever the curve says; a small positional concession near level is a normal move; and an error in a decided position matters less, which the shipped clamps already express. Those are the theses a next round would test, on a fresh sample, with the player's own answers as the key.

6 Reproduction in the lab

The lab recomputes every number of the judge's table from the judge's recorded answers and each run's order key (`tally.mjs`), exactly. The flips are rerun on engine evidence the lab rebuilt on the pinned app (`d8b7712`) from the same games (`reproduce.mjs`); the recorded run used evidence the app wrote before a later change to its review pipeline (review-evidence/4, against review-evidence/5 now), so the scores differ in detail. Table 3 sets the two side by side; the records are read at `b3b827d`.

	recorded	rerun in the lab
engine evidence	review-evidence/4	review-evidence/5
games	60	60
plies with exact scores	3880	3880
graded plies	3578	3578
flips	492 (13.8%)	492 (13.8%)
the player's flips	250 of 1783	247 of 1783
the opponents' flips	242 of 1795	245 of 1795
the copy of the judge matches the app's	yes	yes

Table 3: The flips as recorded and as rerun in the lab.

7 Limitations

- The judge is a model, not the player. The player's own answers on the Calibration page are the key this check is waiting for.
- 25 moves from one player's games is a small sample: it can show that the proposal is not clearly fairer, not that it is clearly worse.
- Only the slope was varied. The judge's reasons suggest that a fairer verdict needs something the curve cannot see, such as whether the move gives away material.